

Long-term Changes in Forest Carbon and Nitrogen Cycling Caused by an Introduced Pest/Pathogen Complex

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ABSTRACT

Invasion of exotic forest pests and pathogens is a serious environmental problem for many forests throughout the world, and has been especially damaging to forests of eastern North America. We studied the impacts of an exotic pest/pathogen complex, the beech bark disease (BBD), in the Catskill Mountains of New York State, USA. In this region, BBD has caused a decline in the basal area of American beech (*Fagus grandifolia* Ehrh.) over the last 60 years and this decline has been accompanied by an increase in the basal area of sugar maple (*Acer saccharum* Marsh.). We studied the impacts of the BBD on carbon (C) and nitrogen (N) cycling using a series of stands that represented a sequence of disease impact and beech replacement by sugar maple. Our study showed that these long-term changes in tree species composition can lead to important changes in C and N cycling in the

ecosystem, including an increase in litter decomposition, a decrease in soil C:N ratio, and an increase in extractable nitrate in the soil and nitrate in soil solution. Rates of potential net N mineralization and nitrification did not change across the BBD sequence, but the fraction of mineralized N that was nitrified increased significantly. Many of the observed changes in ecosystem function are larger in magnitude than those attributed to climate change or air pollution, suggesting that the impacts of invasive pests and pathogens on tree species composition could be one of the most important factors driving changes in C and N cycling in these forests in the coming decades.

Key words: pest; pathogen; beech bark disease; American beech; Catskill Mountains; calcium; carbon; nitrogen; nutrient cycling; forest; soil.

INTRODUCTION

Invasion of forests by non-native insects and pathogens is a serious problem throughout the world, including Asia (Cognato and others 2005),

Africa (Hurley and others 2007), western North America (Meentemeyer and others 2008), and Europe (Mattson and others 2007). The forests of eastern North America have been especially hard-hit, having experienced repeated invasions of insect pests and pathogens imported from Europe and Asia during the last century (Liebhold and others 1995; Lovett and others 2006). Some have caused ecological collapse of the host tree population throughout its range [for example, the specialist pathogen chestnut blight (*Cryphonectria parasitica*)], whereas others have more subtle effects on tree

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species composition [for example, the generalist defoliator gypsy moth (*Lymantria dispar*) (Barron and Patterson 2008)], and several are still spreading but show the potential to eliminate major species (for example, hemlock woolly adelgid (*Adelges tsugae*) (Ellison and others 2005) and emerald ash borer (*Agilus planipennis*) (Poland and McCullough 2006)].

Although introduced insect pests and pathogens can cause serious short-term disruptions in primary productivity and nutrient cycling in forests, often the most important long-term effect of these invasive organisms is a change in the tree species composition of the forest (Stadler and others 2005; Lovett and others 2006). Non-native pests and pathogens can be very effective at changing tree species composition because they are frequently specialists on particular tree species or genera, they can cause high levels of mortality or damage to the host tree populations, and most remain as long-term components of the ecosystem, continually affecting the host tree species.

We studied the beech bark disease (BBD), a disease complex native to Europe that affects American beech (*Fagus grandifolia* Ehrh.) (Houston 1994). Since its introduction into North America at Nova Scotia around 1890, the BBD has spread south and west, and is currently found in 4 Canadian provinces and 14 U.S. States as far south as North Carolina and as far west as Michigan (USDA-Forest-Service 2009). It may eventually spread throughout the range of American beech, which includes most of eastern North America from Cape Breton Island to eastern Texas. The disease complex involves an insect (the beech scale *Cryptococcus fagisuga*) which penetrates the bark of beech trees, allowing the entry of two species of bark-cankering fungus of the genus *Neonectria* (formerly *Nectria*) (Houston 1994). The bark cankers accumulate and usually send the tree into a protracted demise frequently leading to death of mature trees, although a small percentage of American beech trees appear to be resistant to or tolerant of the disease (Houston 1994; Griffin and others 2003). In our principal study site in the Catskill Mountains of southeastern New York State, over 99% of the beech stems larger than 10 cm DBH (diameter at breast height) have some evidence of the disease, and 61% are severely affected (Griffin and others 2003).

BBD usually proceeds through an area as a wave of initial mortality (the “killing front”), followed by the development of an “aftermath forest” in which the disease becomes endemic to the forest and beech mortality continues but at a slower rate (Shigo 1972). Decline of a tree species usually

results in an increase in its competitors, and in the Catskill Mountains and throughout much of the northeastern U.S., the main competitor of American beech is sugar maple (*Acer saccharum* Marsh.). Three lines of evidence suggest that the BBD has had an impact on the vegetation composition of Catskill forests. First, early surveys of vegetation from pre-settlement records showed beech to be the dominant species, whereas more current surveys show beech to be of secondary importance to its competitor sugar maple (McIntosh 1962; Griffin and others 2003), although this change could be a result of many factors in addition to BBD (Fuller and others 1998). Second, foresters reported high levels of beech mortality due to the BBD in the 1950s, soon after the first appearance of the disease in the area (Zabel and others 1958). Third, in a study of the compositional change of the aftermath forests in the Catskills, we relocated a set of 64 forest plots that were initially sampled in 1969–1970 (after the initial killing front) and resampled them in 2000. Data from that study showed sugar maple to be the chief beneficiary of beech decline (Griffin 2005). In plots that lost beech basal area over the 1970–2000 period, there was a significant increase in sugar maple basal area. Extrapolated to the Catskill forest landscape as a whole, beech basal area declined by 1.34 m²/ha (29%) from 1970 to 2000, whereas sugar maple basal area increased by 0.93 m²/ha (Griffin 2005). Throughout the 1970–2000 period Catskill forests were in the “aftermath” phase of BBD. Because most of the beech mortality occurred during the “killing front” in the 1950s, the magnitude of species replacement Griffin (2005) observed between 1970 and 2000 is likely to be only a small portion of the total forest community change induced by BBD since its arrival in the Catskill region.

Tree species composition can have a significant impact on the nutrient cycles of forest ecosystems, because individual tree species have unique characteristics of growth rates, mycorrhizal associations, nutrient uptake, and litter quality, among other properties, that affect the nutrient cycling at a site (Son and Gower 1992; Finzi and others 1998; Lovett and others 2004; Hobbie and others 2006). In forests of the Northeastern U.S., single-species plots of sugar maple and American beech have been shown to differ in litter quality (for example, lignin concentration is higher in beech than maple), soil carbon:nitrogen (C:N) ratio (typically lower in maple stands), and N cycling, particularly in the production and leaching of nitrate (typically higher in maple stands) (Finzi and others 1998; Lovett and Mitchell 2004; Lovett and others 2004).

Therefore, we hypothesized that the species shift engendered by BBD—from mixed beech-maple to maple-dominated stands—would cause reductions in soil C:N ratio, increased nitrification, and increased N leaching. Because these two species have also been shown to differ in litter decomposition rate (Melillo and others 1982) and preference for ammonium versus nitrate as a source of N uptake (Templer and Dawson 2004), other effects on C and N cycles are possible as well.

To test these hypotheses, we established a series of 19 forest stands representing a progression of beech mortality and tree species change caused by the BBD. The stands range from healthy stands of mixed beech and sugar maple to stands that are currently dominated by sugar maple but formerly had a significant component of beech, as evidenced by beech stumps and downed logs. This range of conditions is present in the Catskill landscape because of the heterogeneity of dispersal of the scale insect and the cankering fungi, and because some beech trees appear to be tolerant of the disease and can remain relatively healthy despite being infected (Griffin and others 2003). In our series of stands, we quantified the impact of the BBD and measured key C and N cycling processes to determine how long-term changes in tree species composition caused by the disease impact the functioning of the forest ecosystem.

METHODS

Field Sites and Plot Selection

The research sites are located in the Catskill Mountains, an area of flat-topped mountains and deeply incised valleys encompassing about 5000 km² in southeastern New York State, USA. The bedrock in the higher elevations (>500 m) is relatively homogeneous, consisting primarily of flat-lying sandstones, shales, and conglomerates of Devonian age (Rich 1934; Stoddard and others 1991), and is overlain by glacial till of variable depth (Rich 1934). Soils of the region are primarily thin Inceptisols of moderate to high acidity (Stoddard and others 1991). They are classified as Lithic Dystrochrepts (loamy, skeletal, mixed, mesic) and they are shallow and moderately to excessively well drained (Tornes 1979). The climate of the area is characterized by cool summers and cold winters. The Slide Mountain weather station at 808 m elevation in the central Catskills has a mean annual temperature of 4.3°C (January mean = −8.5°C, July mean = 16.7°C) and a mean annual precipitation of 153 cm, about 20% of which falls as snow.

The vegetation of the region is primarily second-growth hardwood forests dominated by the northern hardwoods forest type (McIntosh 1972). Major tree species include sugar maple and American beech, plus yellow birch (*Betula alleghaniensis* Britt.), eastern hemlock (*Tsuga canadensis* [L.] Carr.), white ash (*Fraxinus americana* L.), red maple (*Acer rubrum* L.), and red oak (*Quercus rubra* L.), with balsam fir (*Abies balsamea* [L.] Mill.) and red spruce (*Picea rubens* Sarg.) occurring on some of the higher mountaintops.

We identified 31 plots representing various stages of BBD impact on beech-maple stands in the Diamond Notch watershed (Latitude 42.147°N, Longitude 74.274°W) near Lanesville, New York State. Prior to beginning our intensive measurements, we chose 22 plots (from the original 31) that (1) were best matched in terms of basal area, elevation, slope, and rock cover, (2) had minimal composition of species other than beech and maple, and (3) represented a sequence from barely impacted stands to those in which all the beech had died (and were present as stumps or downed logs), leaving only maple remaining in the canopy. After litterfall data were collected, we eliminated three plots from consideration because species other than beech and sugar maple contributed greater than 25% of the litter, so the plots did not conform to criterion (2). Thus, there were 19 plots in the final data set. The plots were 20 × 20 m², with a 10 × 10 m² focal area used for nutrient cycling measurements. All of the plots were located within an area of about 3 × 3 km² and ranged from 560 to 790 m elevation. For each plot we calculated a “BBD score” to quantify the progression of beech decline. To calculate the BBD score for each plot, we first rated each beech tree, snag, and log on the plot on a scale of 1–12, where ratings 1–4 are increasing levels of BBD damage to live trees, 5–8 are ratings of increasing degeneration of standing dead snags, and 9–12 are advancing states of decomposition of downed logs (Table 1) (Hancock and others 2008). The BBD score for the plot is the average rating of all of the beech trees, snags, and downed logs in the plot, weighted by their basal areas. The weighting is done to ensure that the average score reflects the dominant trees in the plot more than the subdominant trees. Calculating basal area for downed logs requires estimating their diameters, and although this is more uncertain for decaying logs than for standing trees, in these plots there was sufficient wood left in either the logs or the stumps of the downed trees for us to make reasonable estimates of their diameters. The plot-averaged BBD scores ranged from 2.0 to 10.3.

Table 1. Description of Beech Tree Classes Used in Determining the Severity of BBD and in Establishing the BBD Score, Catskill Mountains, New York

Class	Description of class
1	Live tree, healthy, no sign of BBD
2	Live tree, scale insect present, bark beginning to crack, canopy >75% intact
3	Live tree, bark heavily cracked, cankering, canopy 25–75% intact
4	Live tree, bark severely cracked, large girdling cankers, canopy <25% intact
5	Standing dead tree with fine branches remaining
6	Standing dead tree with only coarse branches remaining
7	Standing dead snag or severe limb loss
8	Downed log, round bole, bark >25% intact, wood solid and hard
9	Downed log, round bole, bark <25% intact, wood solid and hard
10	Downed log, round bole, no bark, wood solid but soft
11	Downed log, elliptical bole, no bark, wood fragmenting
12	Downed log, elliptical bole, wood completely fragmented

Classes 1–4 describe live beech trees, 5–7 describe standing dead beech trees, and 8–12 describe downed logs.

Field and Laboratory Procedures

Sunlit foliage was sampled with a shotgun in mid-summer 2001. Six foliar samples (each containing 10–15 upper-canopy leaves) per plot were taken. Each sample was either beech or sugar maple foliage, and the six samples reflected the relative amounts of beech and sugar maple in the plot. Samples were returned to the laboratory, dried at 60°C, ground, and analyzed for total C and N using a CN analyzer as discussed in the “[Analytical methods](#)” section.

Fine litter was collected from September to November 2001 and 2002 in baskets lined with fiberglass screen. Litter was collected weekly (or bi-weekly in periods of light litterfall) using eight baskets (each having an area of 0.236 m²) per plot. Litter was sorted by species, dried, and ground prior to analysis for total C and N as discussed below.

Soils were sampled in July 2001 with eight cores (each 6.6 cm diameter) per plot taken to a depth of 12 cm, or to the depth of obstruction by large rocks if it was less than 12 cm. Two coring locations were located randomly in each of the four quadrants of the 10 × 10 m² focal plot. These eight cores were separated into organic (Oe and Oa) and mineral horizons, and were composited into four samples per horizon per plot. The plot mean organic horizon depths ranged from 4.3 to 11.1 cm, and mean mineral horizon depths ranged from 0 (for plots with organic soil directly on rock) to 4.2 cm. Samples were kept cool and returned to the laboratory where they were passed through an 8-mm sieve to remove rocks and large roots. Potential N mineralization and nitrification were measured in

28-day aerobic incubations at 20°C and soil moisture at 60% of field capacity (following Lovett and others 2004). Soils were dried at 60°C, ground and analyzed for total C and N. Because soil Ca status may influence the relative abundance of beech and sugar maple (Page and Mitchell 2008), we also measured total elemental calcium (Ca) in mineral soil samples by X-ray fluorescence as an index of the substrate Ca status of the site (see “[Analytical methods](#)” section). Soil solution was collected using ceramic-cup tension lysimeters (made by Soil Moisture, Inc., two lysimeters per plot) installed in the lower B horizon in eight of the plots that spanned the range of BBD score. Lysimeters were installed in October 2001 and soil solution was collected monthly from March 2002 to February 2003, and chemically analyzed using the methods described below.

Litter decomposition was measured with litter decomposition bags made of fiberglass screen (22 × 22 cm², 1.3 mm mesh size). Freshly fallen litter was collected in mesh screening suspended in the plots in October 2001. The litter was sorted and air dried, then 10 g of mixed beech and maple litter was placed in each bag in the proportion of beech:maple litter measured for the plot. (Litter of other species, which constituted less than 25% of the total litter in the plot, was not included.) Thirty litter bags were deployed in each plot in November 2001, such that the bags for each plot contained only the litter from that plot. Five bags per plot were returned to the lab immediately as initial samples, and the other 25 bags were placed in the plot in 5 randomly-placed blocks of 5 bags each. Each bag was placed in contact with the forest

floor, and was pinned in two places to prevent movement. Bags were collected at five times over 3 years (at 5, 8, 12, 20, and 36 months). At each collection time, one bag was collected from each of the five blocks at each plot, for a total of five bags per plot per collection. After collection, bags were cut open, the litter was removed, and any green plant material or roots that had grown into the bags were discarded. The remaining litter was dried, weighed, ground, and analyzed for C and N content.

Analytical Methods

Nitrate and ammonium concentrations in soil KCl extracts were measured using an Alpkem autoanalyzer in the Cary Institute Analytical Laboratory following the methods of Lovett and Rueth (1999). N concentrations in soil solution were measured using a Lachat Quikchem continuous flow analyzer: ammonium using the phenate method, nitrate using the cadmium reduction method, and total dissolved N using the persulfate oxidation method. Dissolved organic C in soil solution was measured by catalytically-aided combustion with a TOC analyzer (Shimadzu model TOC-V CSN). Total C and N concentrations in foliage, litter, and soils were measured using a LECO CN2000 element analyzer (LECO Corp., St Joseph, Minnesota, USA). Total elemental Ca in mineral soil was measured using X-ray fluorescence in the laboratory of Dr. Richard April of Colgate University. Subsamples of mineral soil were powdered, then fused with lithium tetraborate to produce a glass disk. The glass

disk was then analyzed for Ca using a Philips 2404 X-ray fluorescence spectrometer.

Statistical Analysis

Relationships among variables were assessed through simple linear regressions or multiple linear regressions in SAS version 9 (SAS Institute, Cary, North Carolina, USA). A probability of $P < 0.05$ was designated to determine statistical significance.

RESULTS

The progression of beech mortality reflected in increasing BBD score is characterized by a shift from mixed beech/maple forests to forests dominated by sugar maple, with beech increasingly present as standing dead and downed logs. Increasing BBD score was strongly associated with decreases in beech litter input (Table 2, Figure 1A), increases in sugar maple litter input (Table 2, Figure 1A), decreases in live beech basal area and density (Table 2), and increases in dead beech basal area (Figure 1B). Other vegetation indices showed no significant correlation to the BBD score, with the exception of weakly positive correlation with the basal area of species other than beech and sugar maple (Table 2). Foliar litter input to these plots from tree species other than beech or sugar maple was less than 25% of the total litter and showed no trend with BBD score (Figure 1A, Table 2). The BBD score showed no significant relationship to either the total live basal area of all species (Figure 1B) or total live + dead basal area of beech

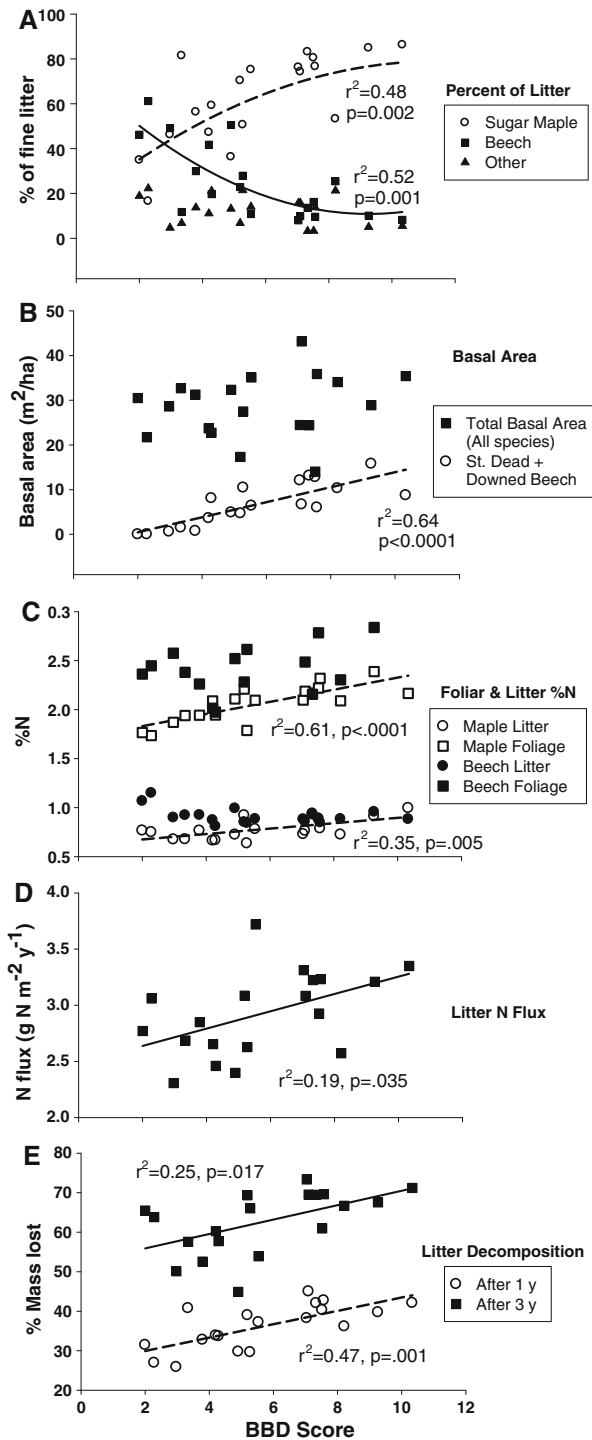
Table 2. Regressions of BBD Score Versus Vegetation Characteristics for the Study Plots

Independent variable	Adjusted r^2	P	Sign of slope
BEE live BA	0.486	0.0005	–
SUM live BA	0.117	0.083	
Other live BA	0.179	0.04	+
Total live BA	0.041	0.408	
BEE standing dead BA	0.093	0.11	
BEE downed logs BA	0.635	<0.0001	+
BEE standing dead + downed logs BA	0.645	<0.0001	+
BEE total live + dead BA	0.002	0.8738	
BEE live stems/ha	0.575	0.0001	–
SUM live stems/ha	0.090	0.1137	
Other live stems/ha	0.050	0.3583	
% SUM in litter*	0.483	0.002	
% BEE in litter*	0.515	0.001	
% Other spp. in litter	0.005	0.376	

BA, basal area; SUM, sugar maple; BEE, beech; Other, all tree species other than beech or sugar maple.

All regressions are linear except those marked with *, which are quadratic. Significant regressions ($P < 0.05$) are in bold. For significant linear regressions, the sign of the slope is indicated.

In all cases $n = 19$ plots.



(Table 2), indicating that the BBD score did not reflect pre-existing variation among plots in these characteristics. The understory (0–10 cm dbh) of the plots was dominated by beech saplings, followed by striped maple (*Acer pennsylvanicum* L.), sugar maple, eastern hophornbeam (*Ostrya virginiana* Mill K. Koch), and yellow birch. The density of beech saplings was negatively correlated with BBD

◀ **Figure 1.** Patterns of vegetation response to BBD impact in the Catskill Mountain region, New York State, as indicated by BBD score. Statistically significant relationships (quadratic in **A** and linear in **B–E**) are indicated by regression lines and regression statistics on the plots; absence of a line indicates non-significant regression. **A** Percent of litter contributed by sugar maple, American beech and other species (dashed line for maple, solid line for beech); **B** total live basal area (stems > 5 cm diameter) and estimated basal area of standing dead (SD) and downed logs of beech; **C** N concentration in litter and foliage of beech and maple (upper regression line is for maple foliage, lower line for maple litter); **D** litterfall N flux; **E** foliar litter decomposition rate expressed as percent litter mass loss after 1 and 3 years.

score ($P < 0.0002$), but the other understory species showed no significant pattern.

N concentration in the green foliage and foliar litter of sugar maple increased with BBD score (Figure 1C), indicating greater availability of N to sugar maple when it is growing with less beech in the stand, even though there was no trend in total live basal area (of all species) with BBD score (Figure 1B, Table 2). There was no significant change in total litter mass or litter C flux with BBD score (Table 3), but there was a significant increase in total litter N flux (Figure 1D), as a result of the change in litter N concentration. The decomposition rate of litter in the plots increased with increasing BBD score (Figure 1E).

There was no significant change in C content of the forest floor and shallow mineral soil, or in forest floor depth (Table 3), but there was a strong decline in forest floor C:N ratio (Figure 2A). The potential N mineralization and nitrification rates in the forest floor showed no significant trends with BBD score (Figure 2B), whereas potential net N mineralization rate in the mineral soil declined significantly (Table 3). A much greater fraction of the mineralized N was nitrified in the plots with increasing BBD score in both organic and mineral soils (Figure 2C). There was a marked shift in the form of extractable N in the soil, which was dominated by NH_4^+ in plots of low BBD score and by NO_3^- in plots of high BBD score (Figure 2D), although it should be noted that this measurement was made at a single point in time and extractable N pools in the soil can be temporally variable. Nitrate concentrations in B-horizon soil solution increased significantly with BBD score (Figure 2E). There was no significant trend in soil solution concentration of SO_4^{2-} with BBD score, therefore the elevated NO_3^- caused an increase in total acid anion (NO_3^- plus SO_4^{2-}) concentration and a shift from SO_4^{2-} -dominated to NO_3^- -dominated anionic charge

Table 3. Results of Regressions of Various Plot Nutrient Cycling Characteristics (Soil Horizons Listed in Parenthesis If Appropriate) Versus BBD Score for the Study Plots

Dependent variable	Adjusted r^2	P	Sign of slope
BEE litter %N	0.136	0.067	
SUM litter %N	0.349	0.005	+
BEE foliar %N	0.042	0.227	
SUM foliar %N	0.610	<0.0001	+
Litter N flux	0.192	0.035	+
Litter C flux	0.023	0.249	
Litter total mass flux	0.057	0.167	
Litter decomp. % mass loss 1 year	0.471	0.001	+
Litter decomp. % mass loss 3 years	0.253	0.017	+
Soil extractable NO_3/NH_4 (mineral)	0.180	0.039	+
Soil extractable NO_3/NH_4 (Organic)	0.550	0.0002	+
Potential N mineralization (mineral)	0.427	0.001	–
Potential N mineralization (organic)	0.104	0.097	
Potential nitrification (mineral)	0.021	0.558	
Potential nitrification (organic)	0.119	0.081	
Nitrification fraction (mineral)	0.533	0.0002	+
Nitrification fraction (organic)	0.645	<0.0001	+
Soil C/N (mineral)	0.500	0.0004	–
Soil C/N (organic)	0.583	<0.0001	–
Org. horizon C pool	0.003	0.838	
Org. horizon N pool	0.071	0.142	
Forest floor depth	0.043	0.392	
Total soil C pool to 12 cm (mineral + organic)	0.035	0.217	
Total soil N pool to 12 cm (mineral + organic)	0.033	0.220	
Soil solution NO_3	0.600	0.015	+
Soil solution DON	0.004	0.888	
Soil solution NH_4	0.004	0.886	
Soil solution DOC	0.092	0.465	
Soil solution SO_4	0.058	0.277	
Soil solution $\text{SO}_4 + \text{NO}_3$	0.616	0.013	+

Significant regressions ($P < 0.05$) are in bold and the sign of the slope is indicated.

In all cases $n = 19$ plots

SUM, sugar maple; BEE, beech.

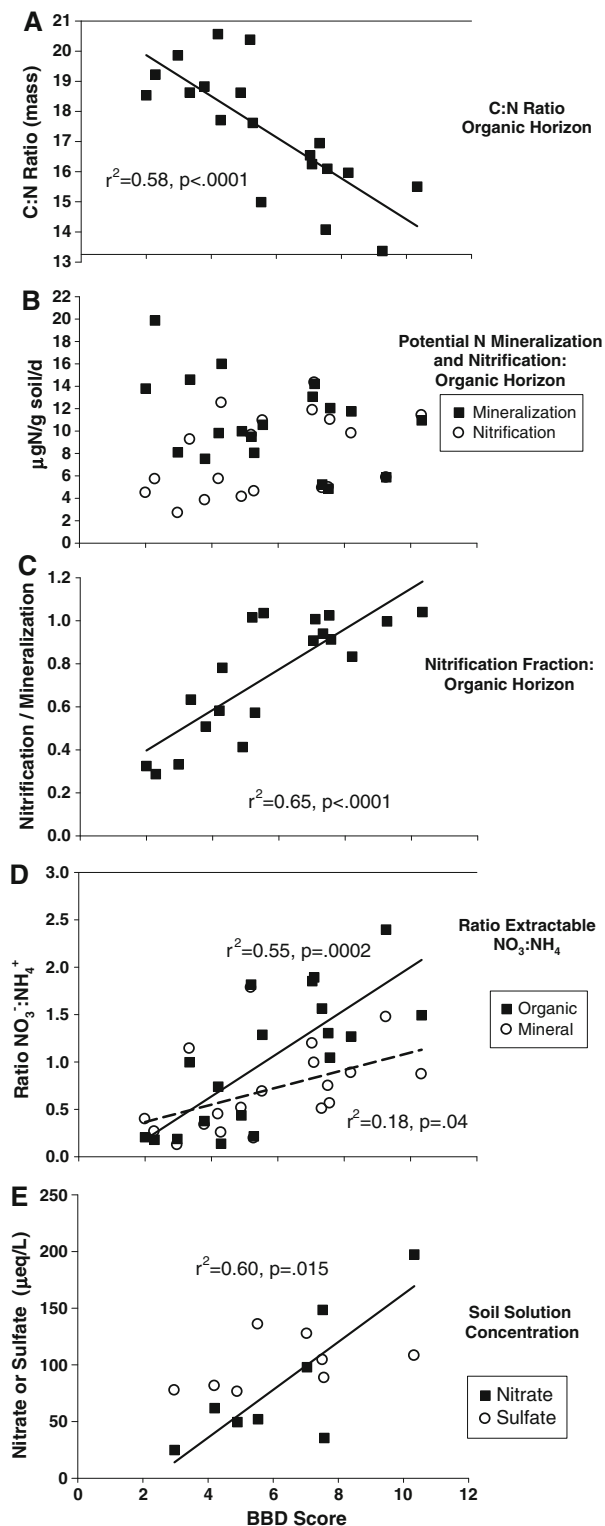
(Figure 2E, Table 3). There were no significant trends in soil solution concentrations of dissolved organic C, dissolved organic N, or NH_4^+ (Table 3).

Our soil sampling revealed that the plots varied in total soil calcium (Ca) as measured by X-ray fluorescence on mineral soil samples. Although there was a tendency for plots with higher BBD score to have higher total Ca, there was no significant correlation between these two variables. To check for a possible confounding effect of total Ca on the response to BBD, we ran stepwise multiple regressions for the response variables reported here using both BBD score and total Ca as independent variables. Total Ca was not a significant explanatory variable for the response variables plotted in Figures 1 and 2, with two exceptions. For soil C:N in the organic horizon (Figure 2A), the BBD score had a partial R^2 of 0.61 ($P < 0.0001$), whereas the

total Ca had a partial R^2 of 0.11 ($P < 0.03$). Likewise, for nitrification fraction in the organic horizon (Figure 2C), the BBD score had a partial R^2 of 0.67 ($P < 0.0001$), whereas the total Ca had a partial R^2 of 0.11 ($P < 0.02$).

DISCUSSION

The results of this study indicate long-term changes in forest C and N cycling resulting from the invasion of the BBD complex. Unlike some other pests and pathogens, the death of beech trees by this disease is not abrupt, but rather occurs gradually as bark cankers progressively reduce the vigor of the tree (Lovett and others 2006). This slow death allows time for neighboring and subdominant trees to overtake the canopy space liberated by the dying trees, minimizing the short-term impacts, such as



reduced productivity and reduced plant uptake, which would be expected from rapid death of trees. For example, infestation of forests by the gypsy moth (a defoliator) and the hemlock wooly adelgid (a phloem-sucking insect that causes extensive tree mortality) are characterized by abrupt changes in N

◀ **Figure 2.** Patterns of soil response to BBD impact in the Catskill Mountain region, New York State, as indicated by BBD score. Statistically significant relationships are indicated by regression lines and regression statistics on the plots; absence of a line indicates non-significant regression. **A** Soil organic horizon C:N ratio; **B** potential N mineralization and nitrification rates in organic horizon soils; **C** nitrification fraction (nitrification rate/mineralization rate) in organic horizon soils; **D** ratio of extractable NO_3^- -N/ NH_4^+ -N in organic and mineral soils; and **E** concentration of NO_3^- and SO_4^{2-} in B-horizon soil solution. The trendline in panel **E** is for NO_3^- concentration; the SO_4^{2-} trend is not significant.

cycling at the time of defoliation or tree death (Webb and others 1995; Jenkins and others 1999; Lovett and others 2002a). If these immediate impacts were occurring in this study we would expect to observe them in stands with BBD scores in the middle of the sequence (that is, 5–7) where the beeches are mainly present as standing dead trees. Similarly, if microclimatic effects associated with the opening of the canopy were important, we would also expect them to be most evident in stands with BBD scores in the middle of the range. However, such patterns are not evident in the data (Figures 1 and 2). Rather, we observed long-term shifts of ecosystem structure and function that reflect the change in tree species composition and its impact on C and N cycling.

The increase in BBD score and the accompanying increase in dominance of sugar maple were associated with a marked decline in forest floor C:N ratio, a strong increase in the fraction of the mineralized N that is nitrified, and elevated levels of extractable NO_3^- in the soil and NO_3^- in soil solution. All of these trends are consistent with our hypotheses. Because these hypotheses were based largely on prior work in single-species stands of beech and maple (for example, Lovett and others 2004), this suggests that basic biogeochemical studies in single-species stands are useful for predicting the impacts of changing species composition in mixed-species stands. High rates of nitrification and NO_3^- leaching have been associated with low forest floor C:N levels in other studies in Europe and North America (Dise and others 1998; Lovett and others 2002b, 2004; Ollinger and others 2002). The increasing ratio of extractable $\text{NO}_3^-:\text{NH}_4^+$ in the soil solution may have an impact on the N nutrition of other trees in the forest, because these trees vary in their preference for either NO_3^- or NH_4^+ as an N source (Templer and Dawson 2004).

The increased foliar decomposition rate with increasing BBD score likely occurred because the

plots with higher BBD scores had a greater percentage of sugar maple litter, which has a lower lignin content (Lovett and others 2004) and is generally more decomposable than beech litter (Melillo and others 1982). The relationship between percent sugar maple in the litter and percent mass loss in the decomposition bags was strong after 1 year of decomposition ($r^2 = 0.77$) but much weaker after 3 years ($r^2 = 0.16$), indicating that the species control over decomposition rate may decline over time. Alternatively, it is known that some litter mixtures show non-additive decomposition rates (that is, the decomposition rate of the mixture cannot be predicted as a linear combination of the rates for the component species) (Gartner and Cardon 2004; Hattenschwiler and others 2005) and it is possible that these non-linear mixing effects, though not evident in our year 1 data, could increase over time.

In addition to the patterns of decomposition, nitrification, and soil solution NO_3^- , we observed other trends that are not so readily explained. For instance, the increased N concentration of sugar maple foliage and litter with increasing BBD score indicates that more N is available to maple trees on plots where beech is absent, despite the apparent lack of a trend in potential N mineralization in the organic soil and an apparent decline in N mineralization in the mineral soil. The reason for the trend in foliar N is unclear from our data, but suggests strong belowground competition for N between these two species in the stands where they co-occur, or perhaps an influence of beech on N availability through microbial processes not captured by our net mineralization assay.

With no change in litterfall C flux and an increase in litter decomposition rate, one might expect to observe a decline in forest floor mass, but we saw no significant change in forest floor depth, mass, or C content associated with BBD score. This may be because soil pools are quite spatially variable and difficult to sample. A previous study (Hancock and others 2008), using a subset of these plots, found no change in ANPP across this disease sequence, but reported a significant decrease in soil CO_2 efflux with increasing BBD score. The lack of change in ANPP is not surprising because the host tree (beech) and its competitor (sugar maple) are both late-successional, shade tolerant trees with similar growth rates. However, the change in soil CO_2 efflux, combined with the increase in litter decomposition rate that we found in this study, suggests substantial changes in soil C dynamics. These decomposition and soil C effects may be mediated by foliar litter quality, in that beech foliage tends to have substantially higher

lignin concentrations and lignin:N ratios than does sugar maple foliage (Lovett and others 2004). However, other influences are also possible, including differences in root tissue chemistry, root exudation, and soil microbial communities, including mycorrhizal symbionts. For example, maple has arbuscular mycorrhizae (AM), whereas beech is dominated by ectomycorrhizae (EM). Compared to AM, EM tend to maintain a greater mass of hyphae, but EM hyphae are typically more difficult to decompose (Langley and Hungate 2003). Another possibility is differences in microclimate or soil moisture caused by the species change. Sugar maple is known to promote water redistribution in the soil through "hydraulic lift," which may keep surface soils more moist (Dawson 1993), possibly enhancing decomposition and other soil microbial activity.

The species change resulting from this disease causes a major shift in the amount and chemistry of acid anions in soil solution in these stands. Because the lysimeters were below the predominant rooting zone of the trees, the concentrations of these ions in the soil solution is probably a good index of the leaching of the ions from the plots. Nitrate leaching is less important than SO_4^{2-} leaching on a charge equivalent basis in watersheds throughout the Catskills as a whole (Lovett and others 2000), but soil solution NO_3^- is equal to or larger than SO_4^{2-} in the high-BBD-score plots in this study. The potential increase in NO_3^- leaching and total anion leaching is important because the Catskill region, like many areas of eastern North America, is subject to soil and water acidification caused by acid deposition (Lawrence and others 1999) and because the streams of this region drain to several eastern U.S. estuaries (Chesapeake Bay, Delaware Bay, Long Island Sound) that are experiencing eutrophication from excess N (Scavia and Bricker 2006).

Although we do not know the exact dates when the BBD first affected the plots, we know that the BBD score quantifies a process of beech mortality and tree replacement that typically requires decades to occur. However, the BBD score is only a rough gauge of time for several reasons. For the first four BBD classes in Table 1, the severity of BBD impact in live trees is a function not only of time, but also of the rate of progression of the disease, which can be influenced by site conditions and the resistance of the trees to the disease. For BBD classes 5–12, the BBD scores quantify increasing deterioration of standing dead trees and increasing states of decomposition of downed logs, which can be influenced by site conditions such as windiness and moisture. Thus, over the entire range of BBD scores, the increasing score reflects the passage of time as stands

pass through the various stages in sequence. However, individual stands may pass through these stages at different rates, imparting variability in the relationship between BBD score and time. Based on the relationship between the length of time that a downed log has been on the ground and its decay condition in similar forests (Arthur and others 1993), we estimate that the downed logs in our highest BBD-score plots have been on the ground for 20–30 years. Because the BBD may take 5–10 years to kill a tree, we infer that this entire disease impact sequence, from infection of beech to its replacement by maple, typically spans three decades or more.

Because this study is a space-for-time-substitution, our conclusions are subject to the uncertainty in controlling plot characteristics other than BBD. By considering 31 potential plots and choosing the best-matched set of 19 (based on slope, elevation, and vegetation) for measurement, we tried to minimize this uncertainty. The BBD score was strongly correlated with vegetation indices associated with decline of beech and replacement by sugar maple, and not correlated with other vegetation measures. There was no significant correlation between BBD score and elevation or slope of the plots. However, we found that there was a significant effect of slope aspect, in that the plots on the west side of this watershed had higher BBD scores than plots on the east side. This observation leads to the questions of why the BBD scores are higher on the west side compared to the east, and whether the factor that causes the difference also influences the response variables and potentially confounds the use of the BBD score. One possibility is a moisture difference, because the slopes on the west side of the watershed face east and thus receive direct sunlight in a cooler part of the day. If this effect were manifested as a canopy microclimate difference that made conditions on east-facing slopes more conducive to progression of the disease, this would simply be one factor contributing to the variability in BBD impact we observe across the landscape, and would probably not confound the response variables we measured. If the microclimate effect was manifested as difference in soil moisture between east- and west-facing slopes, it could potentially confound the use of the BBD score. We did not monitor soil moisture in these plots, but at the time of our soil sampling for the N mineralization assay, there was no significant difference in field moisture between the east and west sides of the watershed, in either the organic or mineral horizons. Moreover, beech tends to be somewhat more drought-sensitive than sugar maple (Caspersen and Kobe 2001), which should make beech more competitive on the west side of the watershed, the

opposite of the pattern observed. Another factor that could account for the aspect difference in BBD scores is soil Ca status, because sugar maple is known to be more Ca-demanding than beech (Kobe 1996; Christopher and others 2006), and we found that the total soil Ca on the west side of the watershed was, on average, about 50% higher than on the east side, though the difference was not statistically significant (east side mean (SE) = 0.11% (0.04) CaO, west side mean (SE) = 0.17% (0.03), $P = 0.18$). This difference suggests that there may be an interaction between Ca status of the soil and the ability of sugar maple to colonize the site after the BBD impact. We are currently evaluating this hypothesis across a broader geographic scale in the northeastern U.S. For this study, our evaluation of the effect of soil Ca on the C and N cycling variables we measured indicates that Ca is not a major confounding factor, but it may have a secondary influence on soil C:N ratio and nitrification fraction. We cannot rule out that some other characteristics of the east versus west side of the watersheds could have affected the C and N cycling patterns we measured.

To our knowledge, this is the first study to document the long-term (decadal) changes in forest C and N cycling caused by an introduced forest pest. BBD has spread throughout the core of the range of American beech in the northeastern U.S. and eastern Canada and is continuing to spread into the southeastern and midwestern U.S. (Morin and others 2007). Whether beech retains its dominance in these affected forests, as well as the identity of the species that replace it if it does not, will vary throughout the range (Houston 1975; Twery and Patterson 1984; Forrester and others 2003). Other pests currently spreading in the eastern U.S. also have the capability of changing species composition over large areas, for example, the hemlock woolly adelgid and the emerald ash borer, and may produce comparable long-term effects in ecosystem function. The nature and strength of these effects depends on the properties of the host tree species and those species that replace it, particularly their growth rate, litter chemistry, litter decomposition rate, C allocation, mycorrhizal affinity, and amount and form of N uptake from the soil. These effects are complex but predictable, at least qualitatively, given a general knowledge of the competitive interactions in the forest tree community and some basic information on the physiological properties of the species involved (Lovett and others 2006). Other likely effects of shifts in tree species composition that we did not study, but are potentially important, include changes in habitat and food supply for wildlife and altered timber resources.

The spread of introduced pests and pathogens may be influenced by interactions with other forms of environmental change. For instance, warming temperatures can enhance the spread of some pests (Dukes and others 2009), whereas increased N in plant tissues (such as may occur with elevated atmospheric N deposition) increases the susceptibility of some trees to phytophagous insects (McClure 1991), including the degree of infestation of beech scale on beech trees (Latty and others 2003). Warming temperatures and changing precipitation will interact with changing species composition to affect most of the C and N cycling processes in an ecosystem. Furthermore, prediction and management of the effects of climate change and air pollution will be made more complex by the changing tree species composition caused by pests and pathogens. For instance, forests shifting from beech to sugar maple dominance will have greater NO_3^- leaching to surface waters, potentially reducing the benefits of regulations that curtail N oxide emissions. Similarly, models that predict altered tree species composition under the scenarios of future climate change (for example, Iverson and others 2008) will require revision if major tree species such as American beech undergo pest-induced declines.

Unlike other forms of human-accelerated environmental change affecting these forests, introduced pests and pathogens have the potential to drastically reduce the abundance and importance of major tree species in the next 20–30 years. The magnitude of the changes we observed in litter decomposition, soil C:N ratio, nitrification, and nitrate leaching as a result of the BBD-induced species change is larger than those typically attributed to climate change or air pollution in this region (Murdoch and others 1998; Lovett and Rueth 1999; Driscoll and others 2003; Knorr and others 2005). This study illustrates that, by changing the species composition of forests, introduced pests and pathogens can alter fundamental ecosystem processes over the long term, and may be one of the most important factors driving changes in C and N cycling in these forests in the coming decades.

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